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SIMATS ENGINEERING

DEEP LEARNING ARCHITECTURES – Second Unit Test

Score: 22 / 25

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Answer Details

Q1: Which statement is TRUE regarding deeper neural networks?

A. They always train faster

B. They can learn more complex hierarchical features

C. They require fewer parameters

D. They eliminate overfitting completely

Submitted Answer: They can learn more complex hierarchical features

Correct Answer: They can learn more complex hierarchical features

Q2: Which application commonly uses Autoencoders?

A. Data compression and anomaly detection

B. Database indexing

C. Network routing

D. Compiler optimization

Submitted Answer: Data compression and anomaly detection

Correct Answer: Data compression and anomaly detection

Q3: Unsupervised learning is characterized by:

A. Training with labeled data

B. Training without labeled data

C. Training only convolutional networks

D. Training reinforcement agents

Submitted Answer: Training without labeled data

Correct Answer: Training without labeled data

Q4: Deep Learning is a subset of:

A. Data Mining

B. Artificial Intelligence only

C. Machine Learning

D. Computer Graphics

Submitted Answer: Machine Learning

Correct Answer: Machine Learning

Q5: An Autoencoder consists of:

A. Generator and discriminator

B. Encoder and decoder

C. Visible and hidden layers only

D. Input and output layers only

Submitted Answer: Encoder and decoder

Correct Answer: Encoder and decoder

Q6: Increasing the width of a neural network means:

A. Adding more hidden layers

B. Adding more neurons in each hidden layer

C. Increasing the learning rate

D. Increasing the dataset size

Submitted Answer: Adding more neurons in each hidden layer

Correct Answer: Adding more neurons in each hidden layer

Q7: Which of the following is a real-world application of Deep Learning?

A. Image recognition

B. Speech recognition

C. Medical diagnosis

D. All of the above

Submitted Answer: All of the above

Correct Answer: All of the above

Q8: Representation Learning refers to:

A. Storing data in databases

B. Automatically discovering useful features from data

C. Compressing files

D. Visualizing graphs

Submitted Answer: Automatically discovering useful features from data

Correct Answer: Automatically discovering useful features from data

Q9: Increasing the depth of a neural network means:

A. Increasing the batch size

B. Adding more hidden layers

C. Increasing the number of neurons only

D. Increasing epochs

Submitted Answer: Adding more hidden layers

Correct Answer: Adding more hidden layers

Q10: Which activation function helps reduce the dying ReLU problem by assigning a small slope to negative values?

A. Sigmoid

B. Tanh

C. Leaky ReLU

D. Softmax

Submitted Answer: Leaky ReLU

Correct Answer: Leaky ReLU

Q11: What is the main advantage of ReLU?

A. Produces outputs between 0 and 1

B. Faster convergence and reduced vanishing gradient problem

C. Always differentiable

D. Generates negative outputs only

Submitted Answer: Faster convergence and reduced vanishing gradient problem

Correct Answer: Faster convergence and reduced vanishing gradient problem

Q12: RBMs are commonly used for:

A. Object detection only

B. Feature learning and dimensionality reduction

C. Image compression only

D. Sorting algorithms

Submitted Answer: Feature learning and dimensionality reduction

Correct Answer: Feature learning and dimensionality reduction

Q13: Which of the following is the key advantage of Deep Learning over traditional Machine Learning?

A. Requires less data in all cases

B. Automatically learns feature representations from raw data

C. Uses only linear models

D. Does not require training

Submitted Answer: Automatically learns feature representations from raw data

Correct Answer: Automatically learns feature representations from raw data

Q14: A Restricted Boltzmann Machine (RBM) consists of:

A. Input, hidden, and output layers with full connections

B. Visible and hidden layers without intra-layer connections

C. Only one hidden neuron

D. Multiple output layers

Submitted Answer: Visible and hidden layers without intra-layer connections

Correct Answer: Visible and hidden layers without intra-layer connections

Q15: Which activation function has exponential behavior for negative inputs?

A. ReLU

B. Leaky ReLU

C. ELU

D. Softmax

Submitted Answer: ELU

Correct Answer: ELU

Q16: Which Deep Learning application is widely used in autonomous vehicles?

A. Payroll management

B. Image classification and object detection

C. Text editing

D. Spreadsheet analysis

Submitted Answer: Text editing

Correct Answer: Image classification and object detection

Q17: ELU stands for:

A. Exponential Learning Unit

B. Exponential Linear Unit

C. Extended Linear Unit

D. Exponential Logistic Unit

Submitted Answer: Exponential Linear Unit

Correct Answer: Exponential Linear Unit

Q18: The dying ReLU problem occurs because:

A. ReLU produces large outputs

B. Some neurons permanently output zero

C. ReLU is computationally expensive

D. ReLU cannot process images

Submitted Answer: Some neurons permanently output zero

Correct Answer: Some neurons permanently output zero

Q19: An Autoencoder is mainly designed to:

A. Classify images

B. Learn compressed representations of input data

C. Detect network attacks only

D. Perform regression tasks only

Submitted Answer: Learn compressed representations of input data

Correct Answer: Learn compressed representations of input data

Q20: Which of the following is an unsupervised neural network model?

A. CNN

B. RNN

C. Restricted Boltzmann Machine (RBM)

D. Perceptron

Submitted Answer: RNN

Correct Answer: Restricted Boltzmann Machine (RBM)

Q21: Leaky ReLU differs from ReLU because it:

A. Eliminates all negative values

B. Allows a small non-zero gradient for negative inputs

C. Uses exponential functions

D. Produces outputs only between -1 and 1

Submitted Answer: Allows a small non-zero gradient for negative inputs

Correct Answer: Allows a small non-zero gradient for negative inputs

Q22: The Rectified Linear Unit (ReLU) activation function is defined as:

A. $f(x) = x^2$

B. $f(x) = 1 / (1 + e^{-x})$

C. $f(x) = \max(0, x)$

D. $f(x) = \tanh(x)$

Submitted Answer: $f(x) = \max(0, x)$

Correct Answer: $f(x) = \max(0, x)$

Q23: Compared to ReLU, ELU:

A. Produces only positive outputs

B. Produces smooth negative values that help learning

C. Has no negative outputs

D. Is slower because it is linear

Submitted Answer: Has no negative outputs

Correct Answer: Produces smooth negative values that help learning

Q24: Which type of learning is used to train an Autoencoder?

A. Reinforcement learning

B. Supervised learning with class labels

C. Self-supervised/unsupervised reconstruction learning

D. Transfer learning only

Submitted Answer: Self-supervised/unsupervised reconstruction learning

Correct Answer: Self-supervised/unsupervised reconstruction learning

Q25: Deep Learning is a subset of:

A. Data Mining

B. Artificial Intelligence only

C. Machine Learning

D. Computer Graphics

Submitted Answer: Machine Learning

Correct Answer: Machine Learning